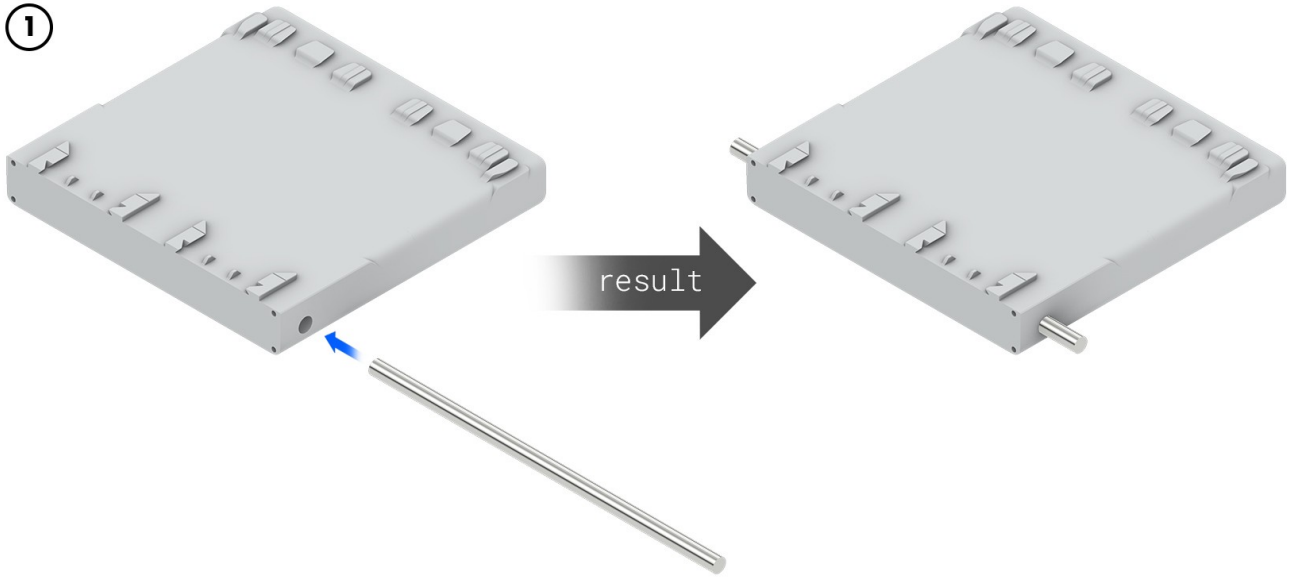
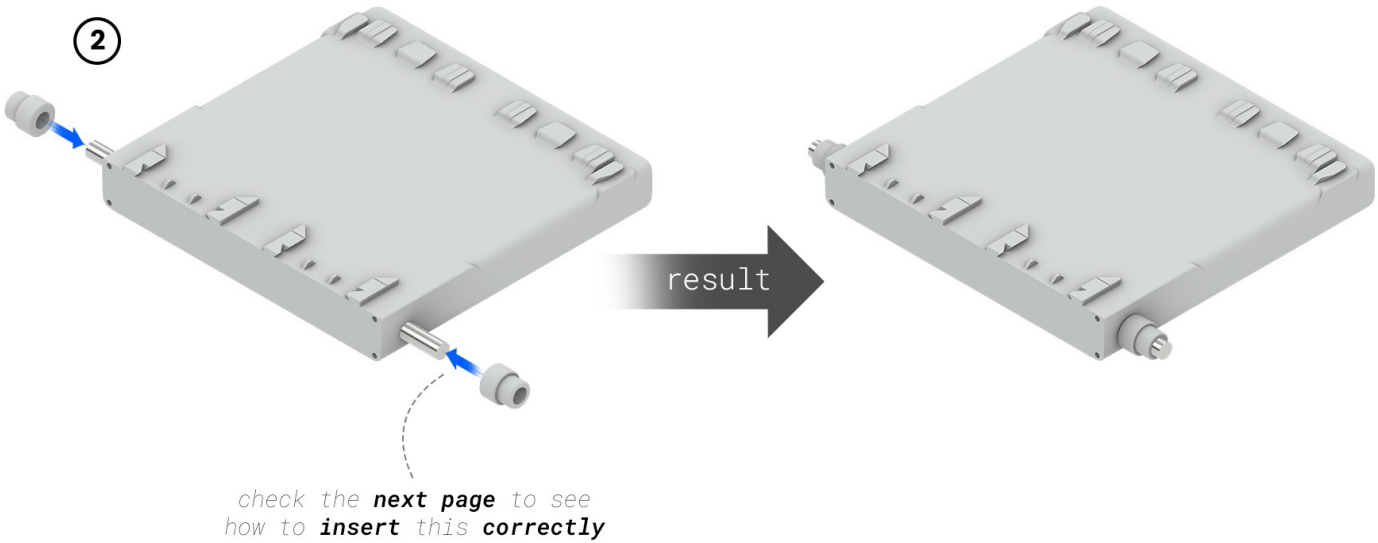


Tower made from L & S modules of different heights; you can use wide L-width backbone parts or universal parts.

Modifications Exploded View & Parts



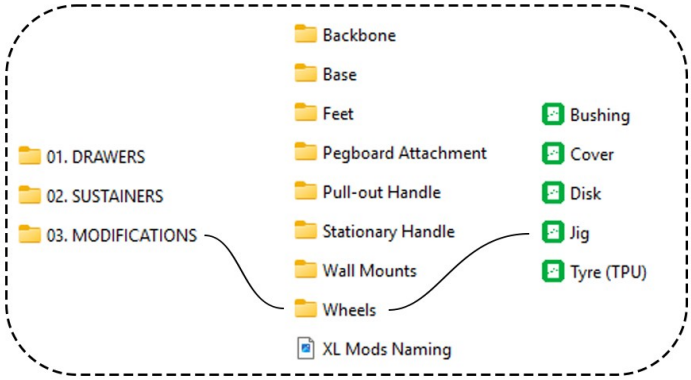


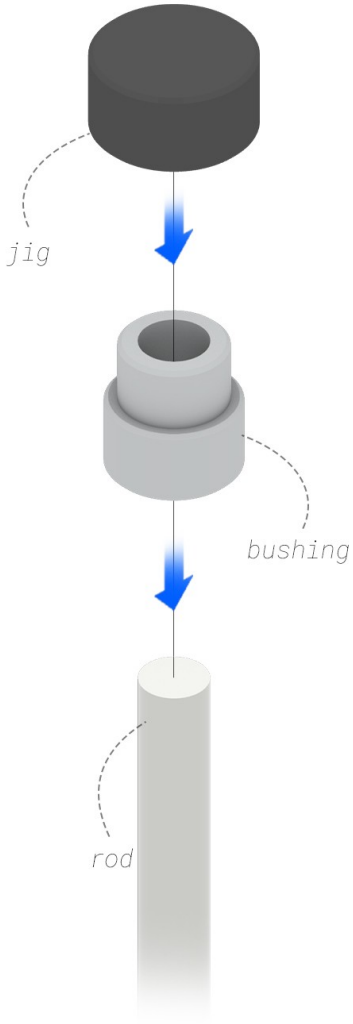
*check the **next page** to see how to **insert** this **correctly***

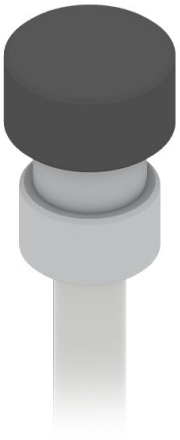
1. cut a rod to 504MM for XL, or 414MM=L, 338MM=M and 248=S, and push it into a base
2. use a special jig to push the bushings from both sides on a rod *(see next page)*

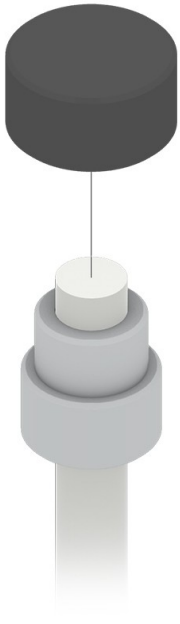
Modifications Assembly 3/15

here's how to find jig



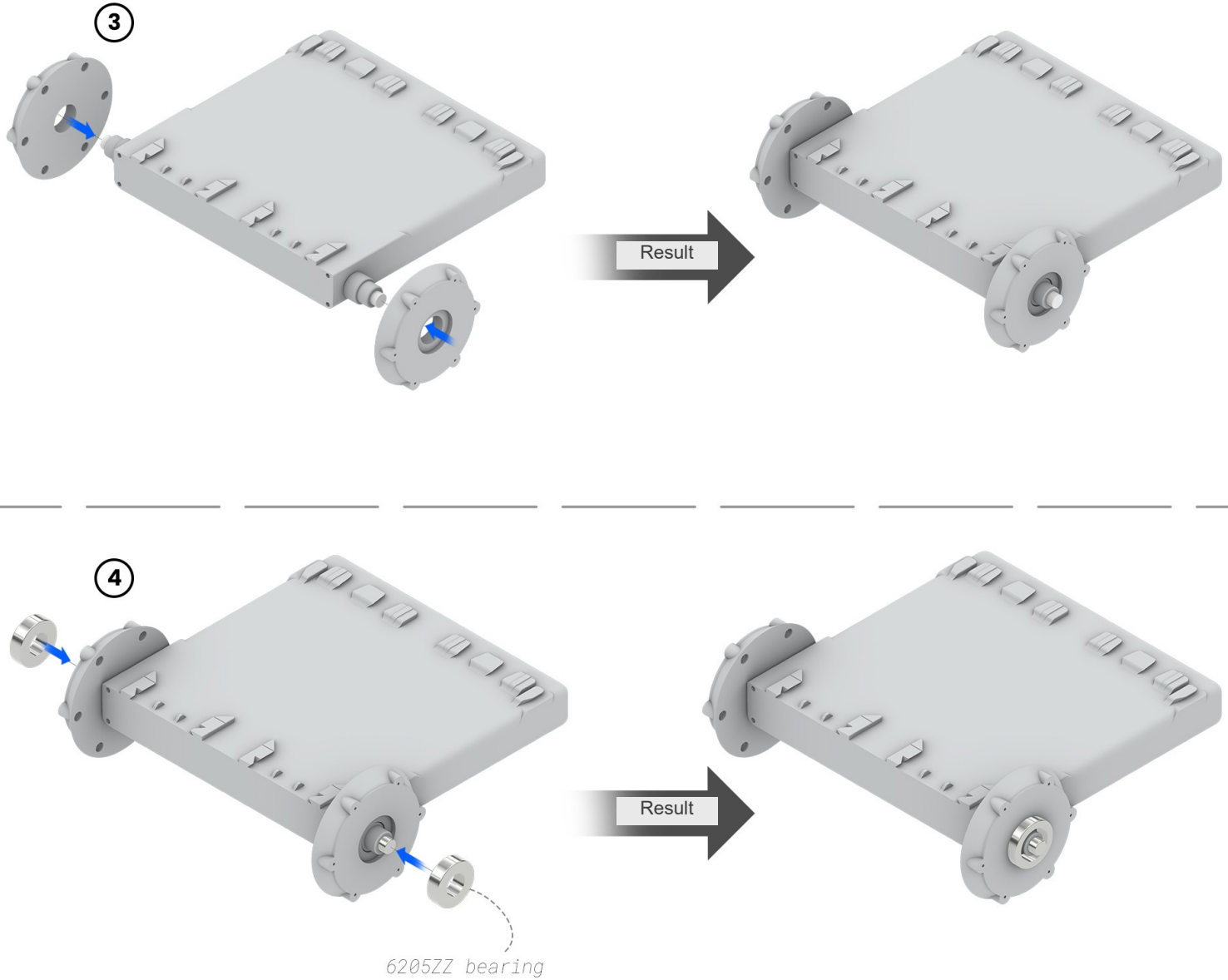




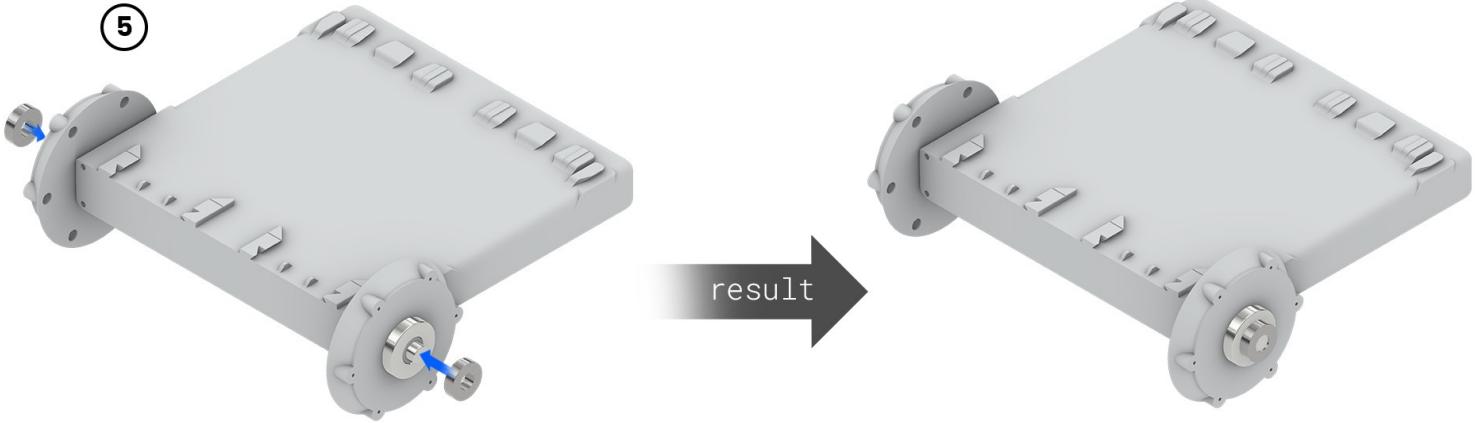


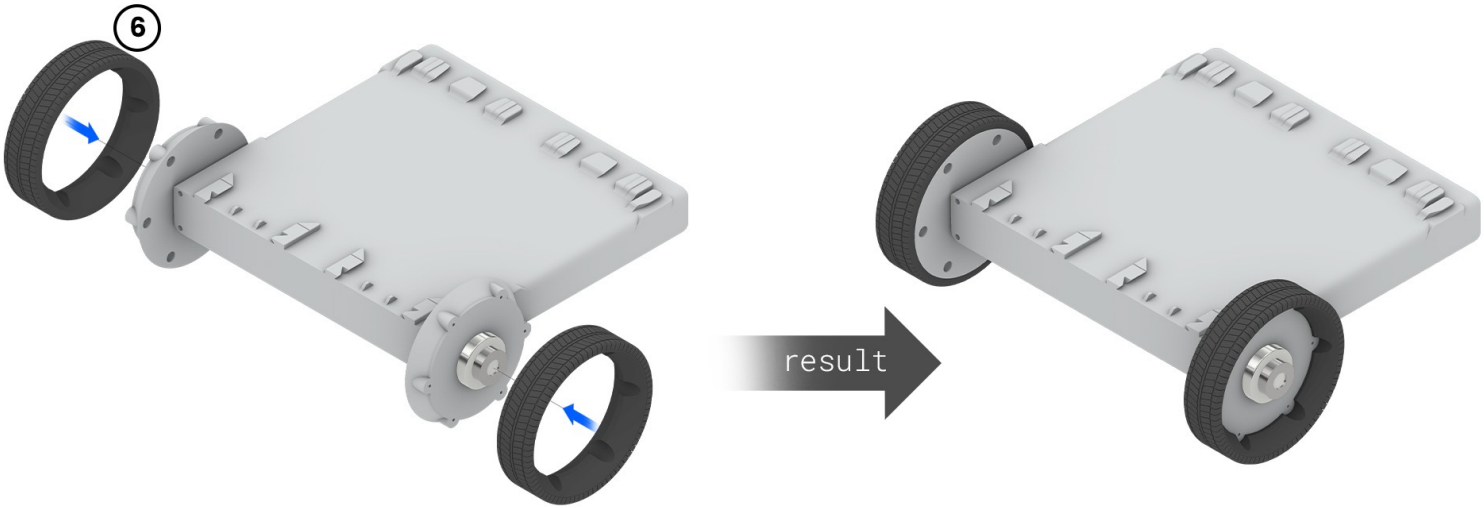
1 **2** **3**

1. Push bushing in first, then jig to leave 10mm rod sticking out; 2. Check alignment; 3. Take out the jig.

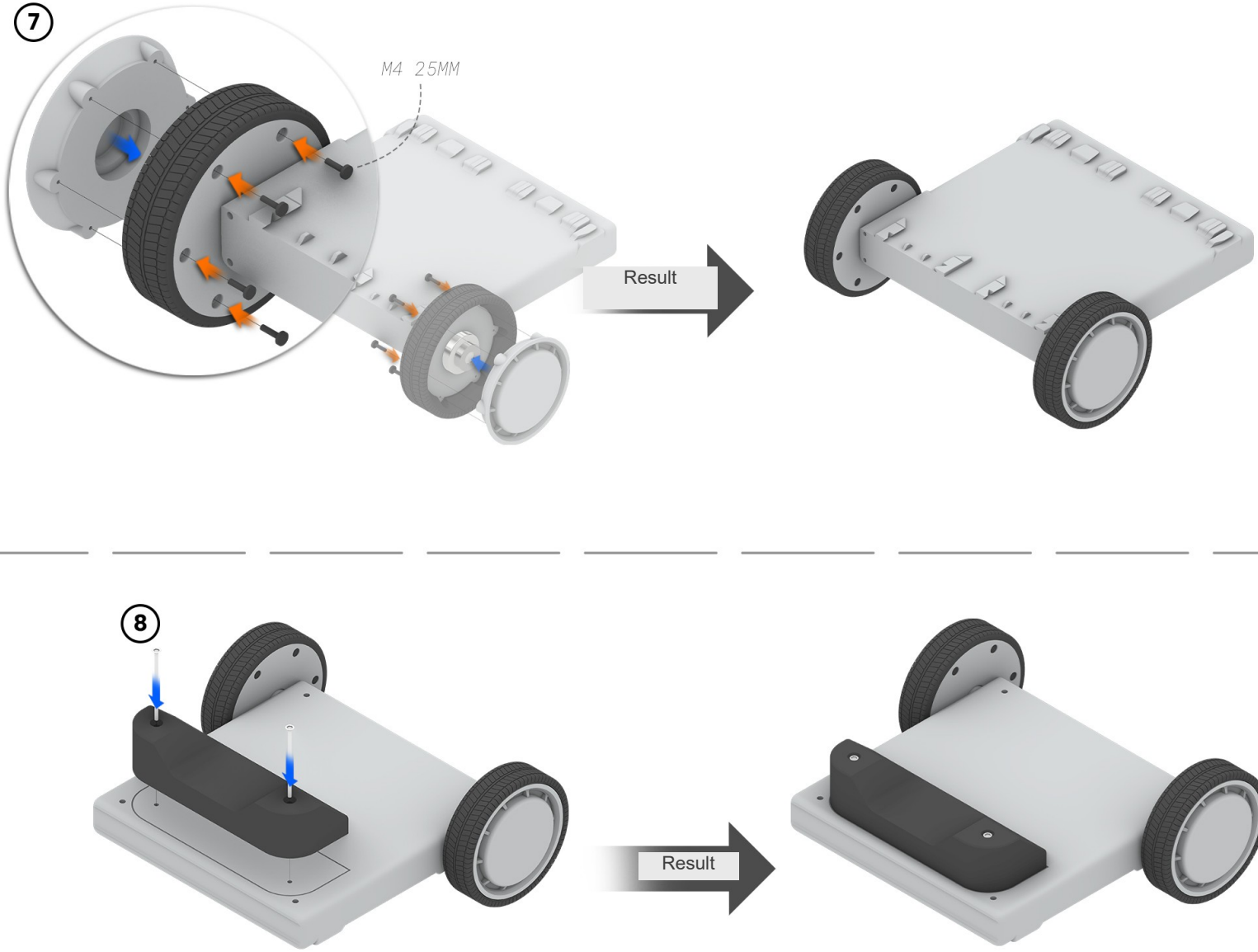


3. Place disks from both sides onto bushing; 4. Push 6205ZZ bearings (25d-52D-15mm) onto printed bushings.

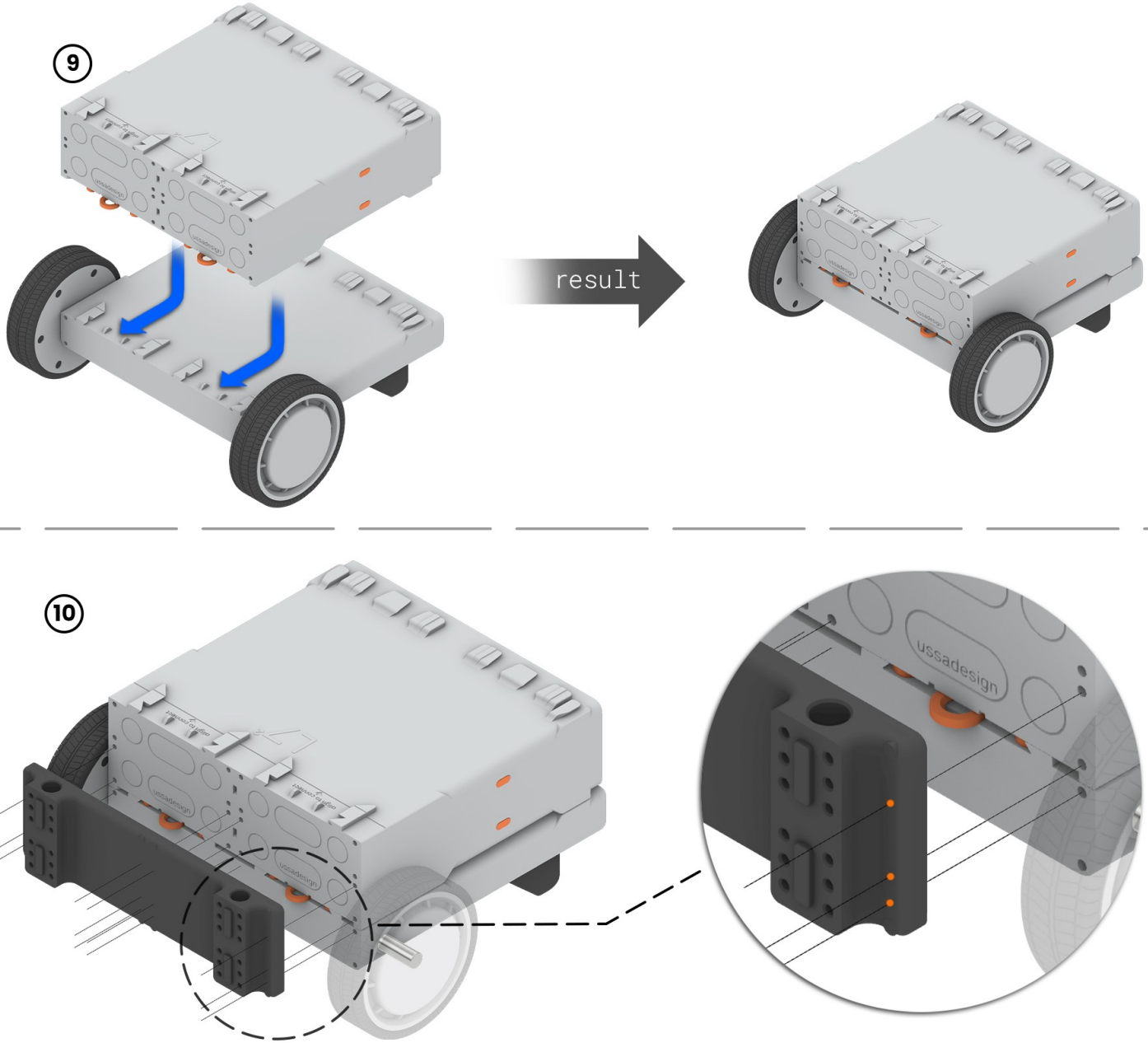




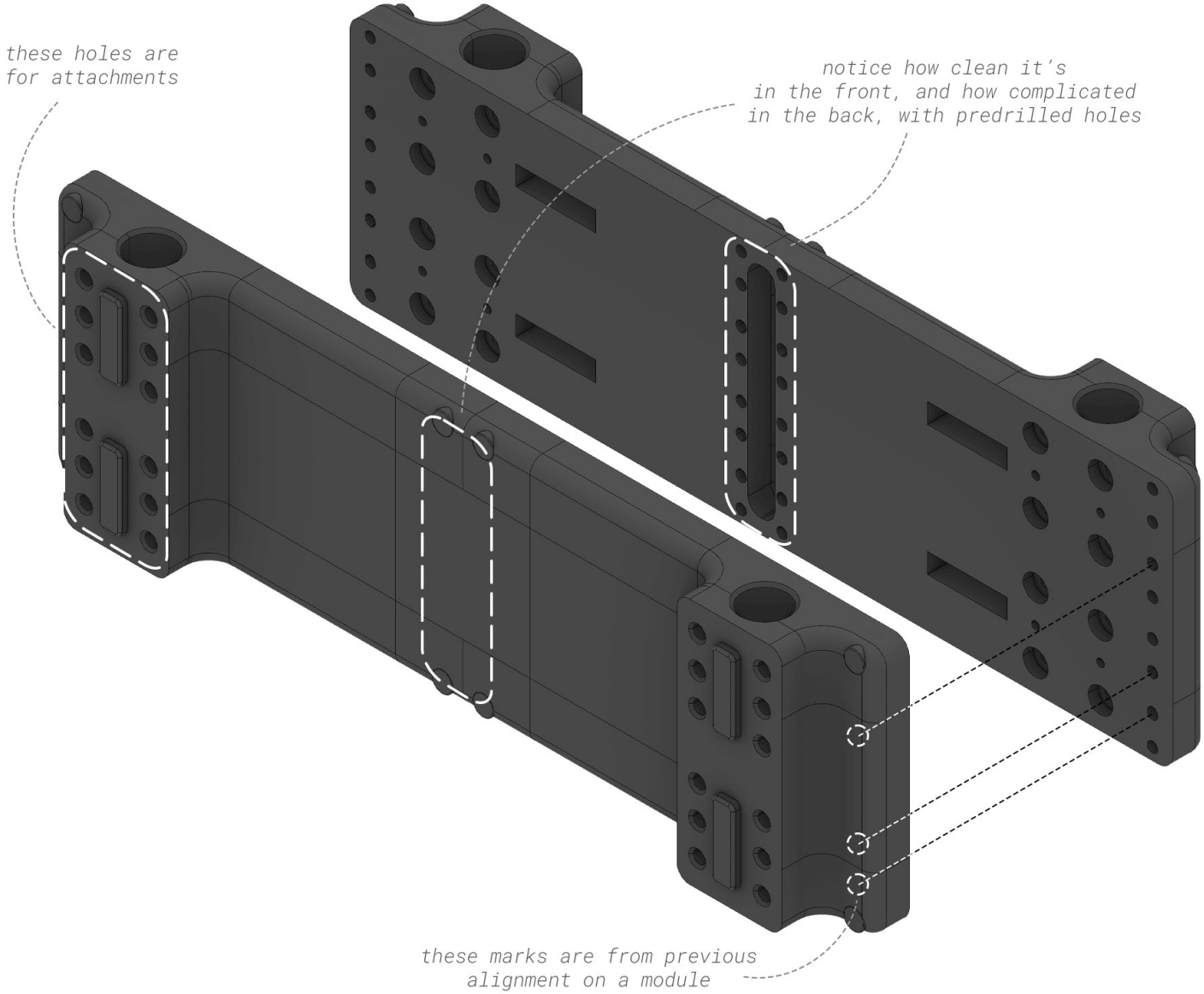
5. Secure bearings with SCSAWs (16d, 35D, 10mm); 6. Recommend TPU for tires; slide them onto disks after printing.



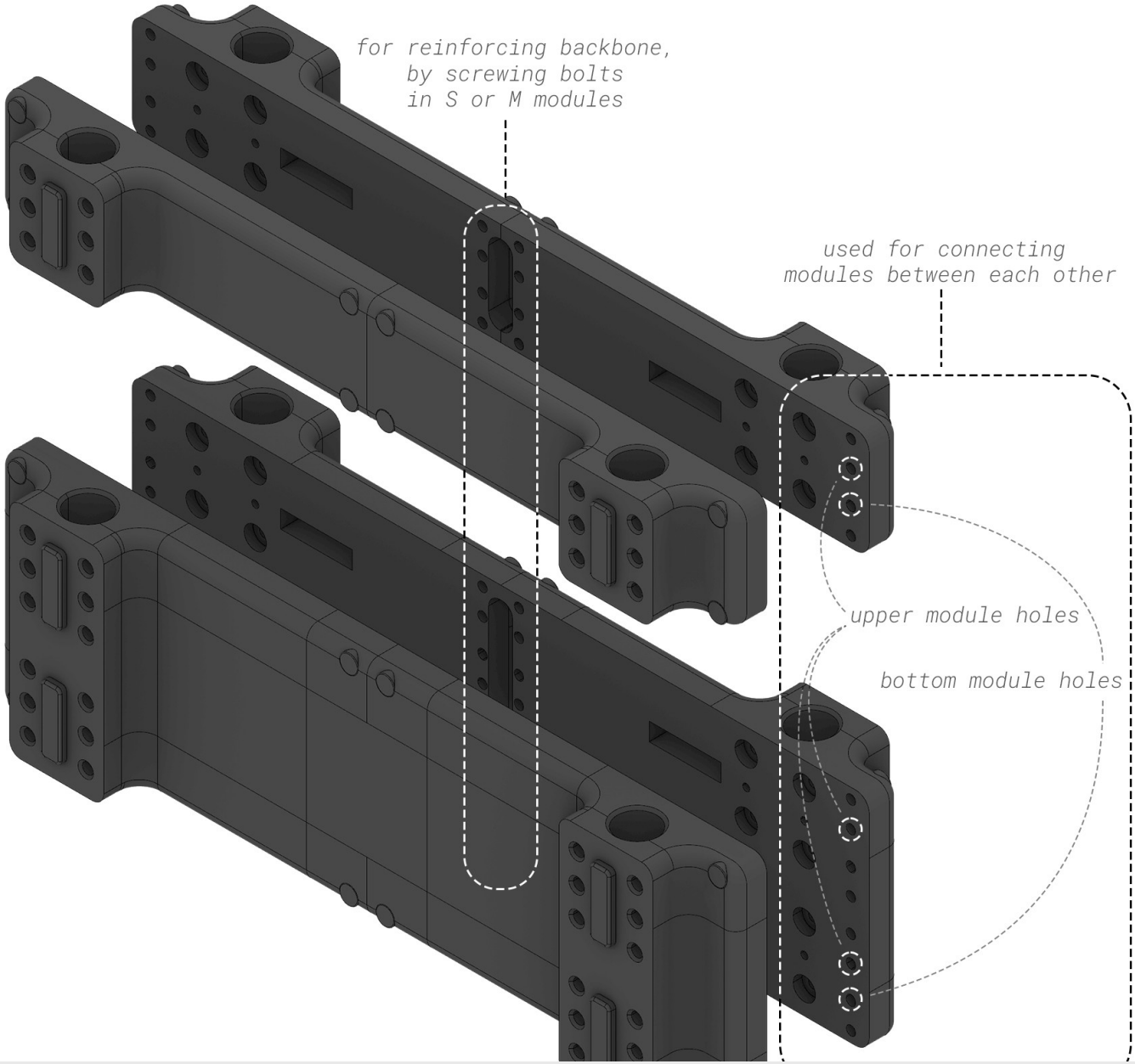
7. Close up with cover using 6× M4 25mm bolts per wheel; 8. Flip upside down and attach foot with M6 90mm.



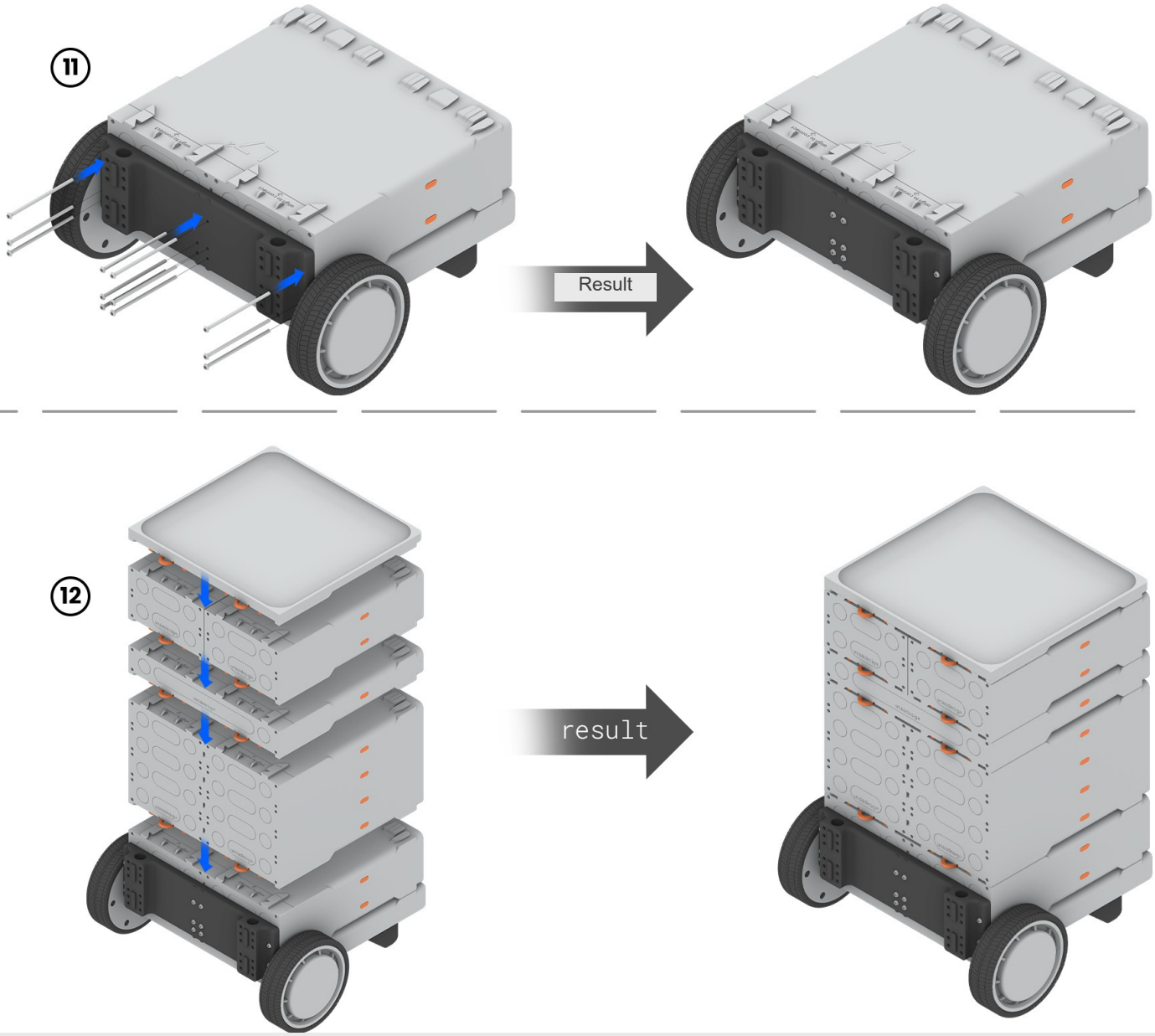
9. HIGHLY RECOMMEND 2F MODULE AFTER BASE; 10. Prepare sealed 2F backbone for drilling; mark the 3 spots.



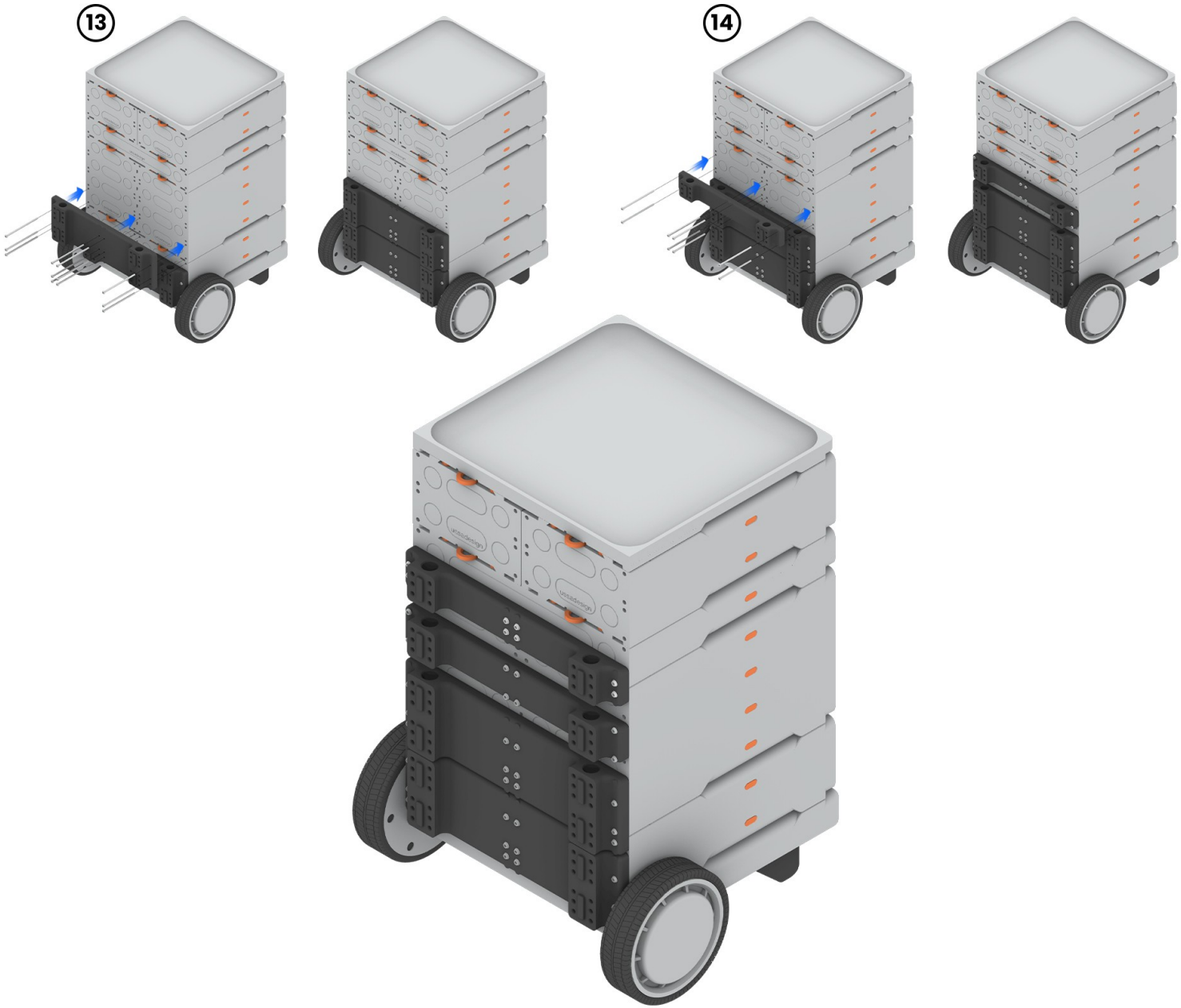
11. After marking spots, drill through with a 4mm bit; notice clean predrilled alignment marks on modules.



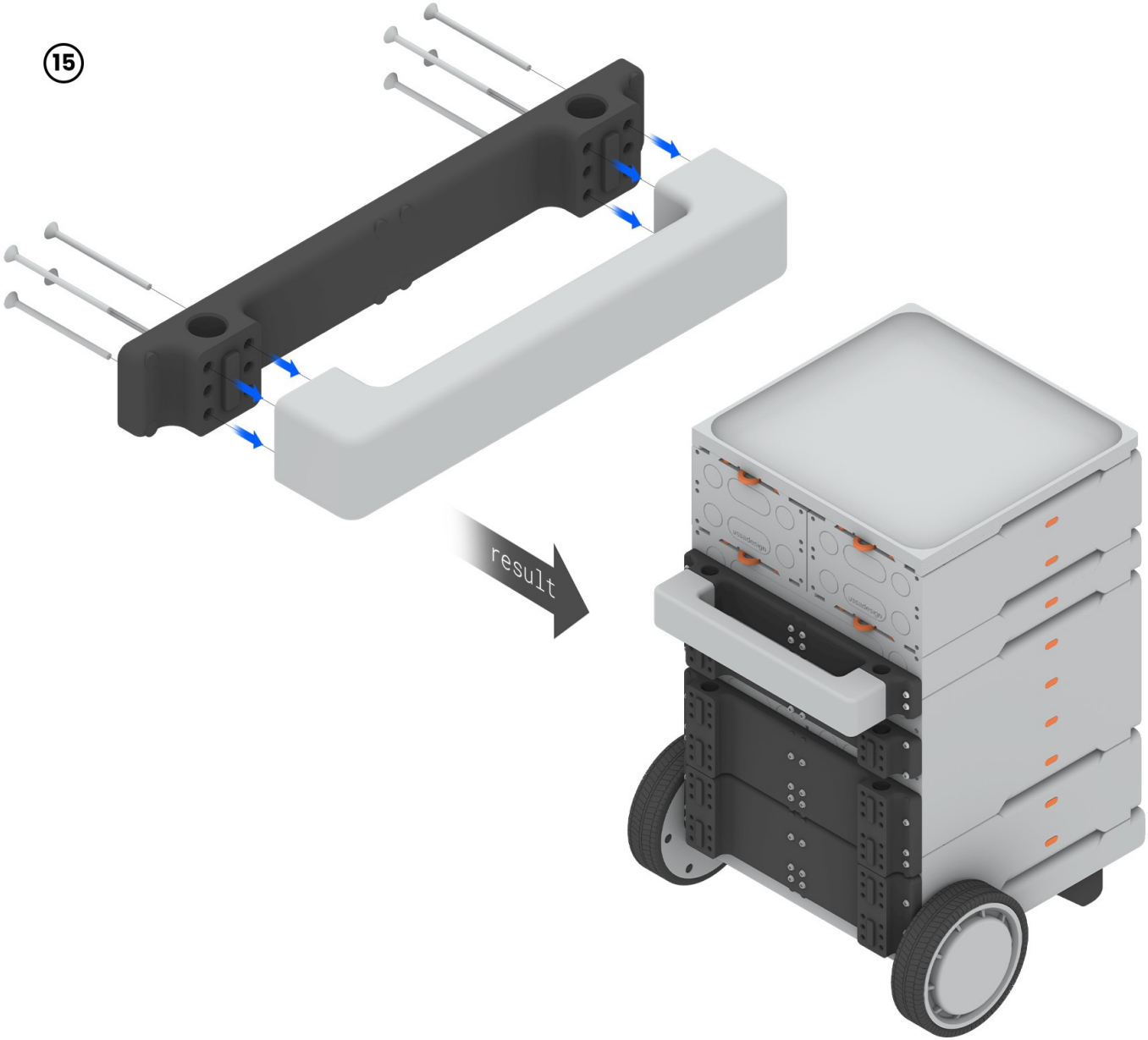
Used for connecting modules between each other and reinforcing backbone by screwing bolts into modules (upper / bottom module holes).



12. Attach backbone to both base and module; HIGHLY RECOMMEND ENDING WITH A 2F MODULE BEFORE THE ROOF.

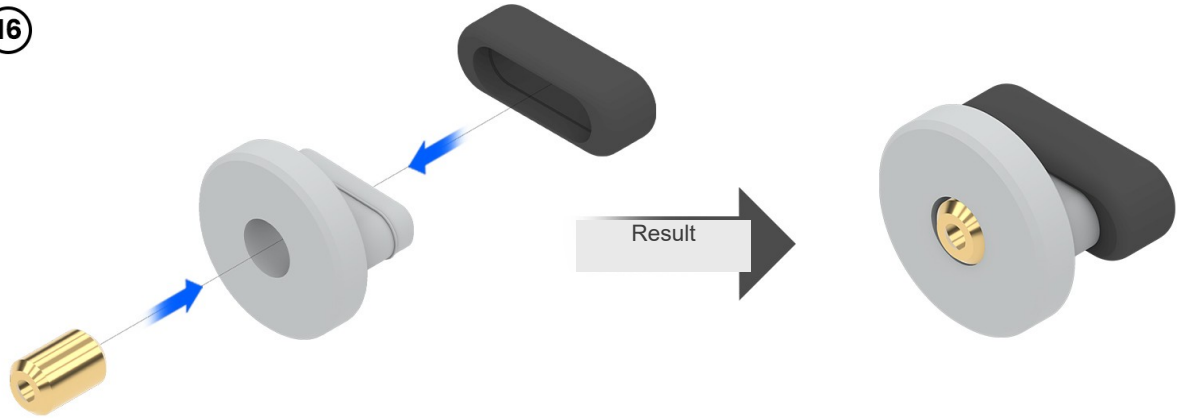


13 & 14. Connect remaining modules with each other; 1F backbone can be used for linking modules as well.

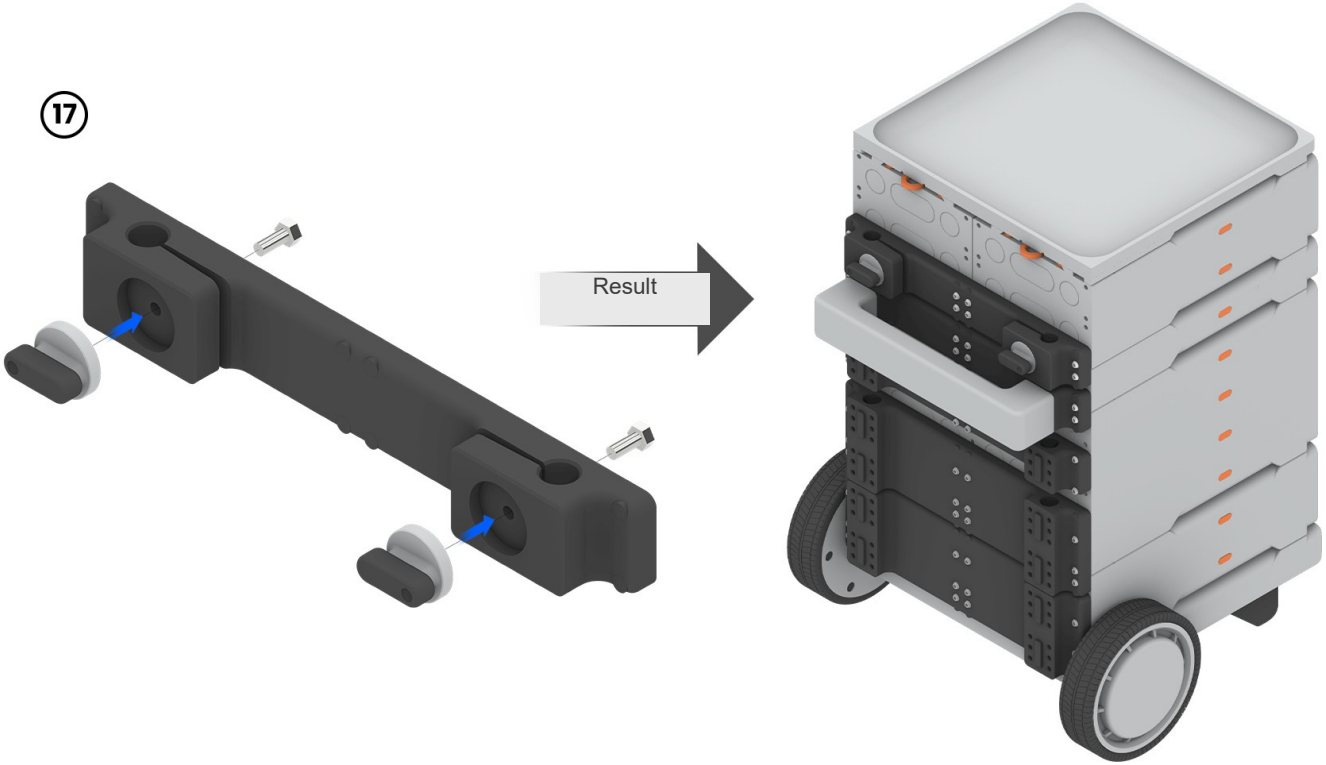


15. If using stationary handle, attach it from the other side of backbone BEFORE attaching backbone to module.

16



17



16. Heat up insert and push straight into knob, then cap it; 17. Use Hex M4 35mm bolt from other side to tighten.

18



result



19

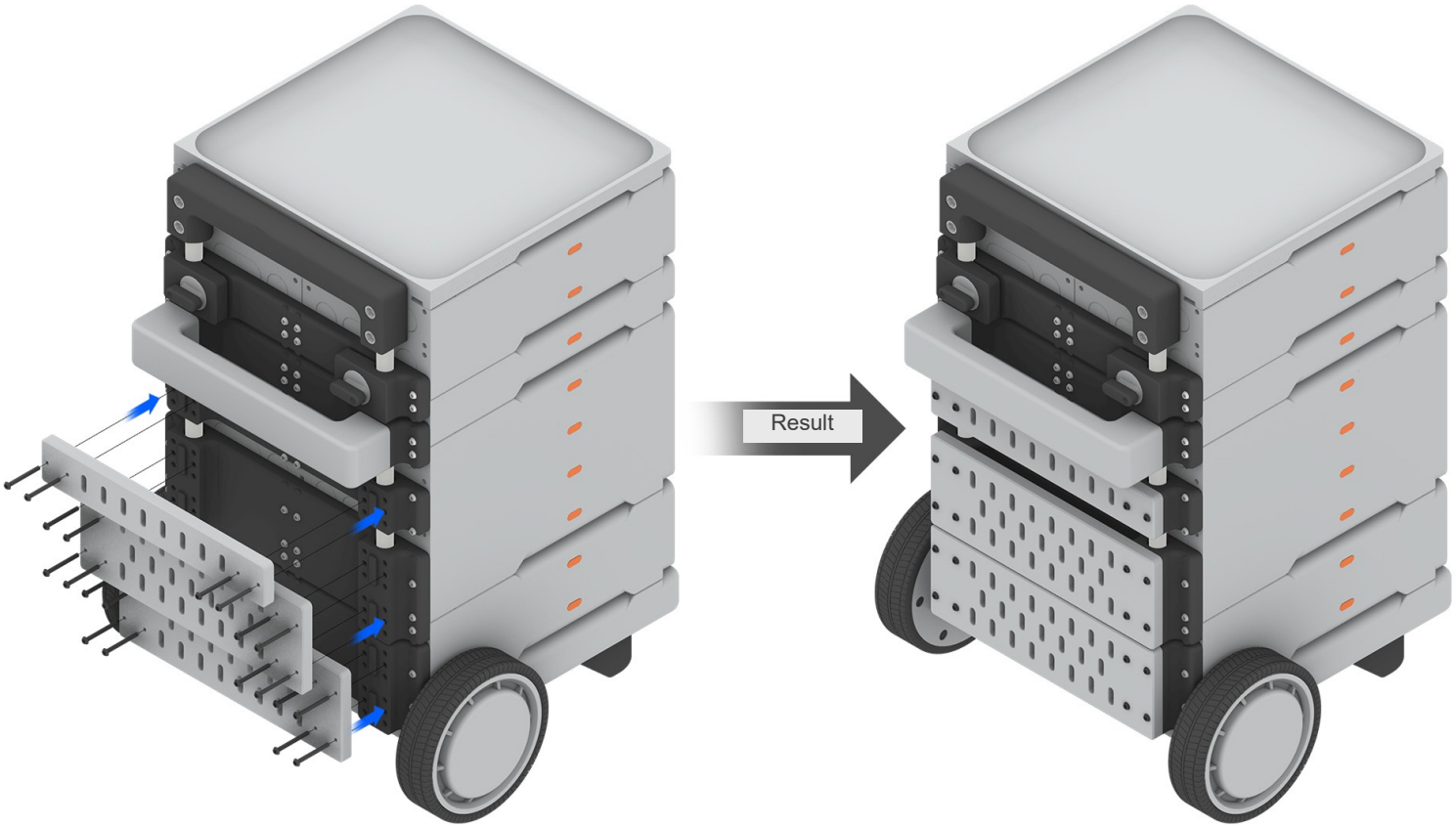


Result



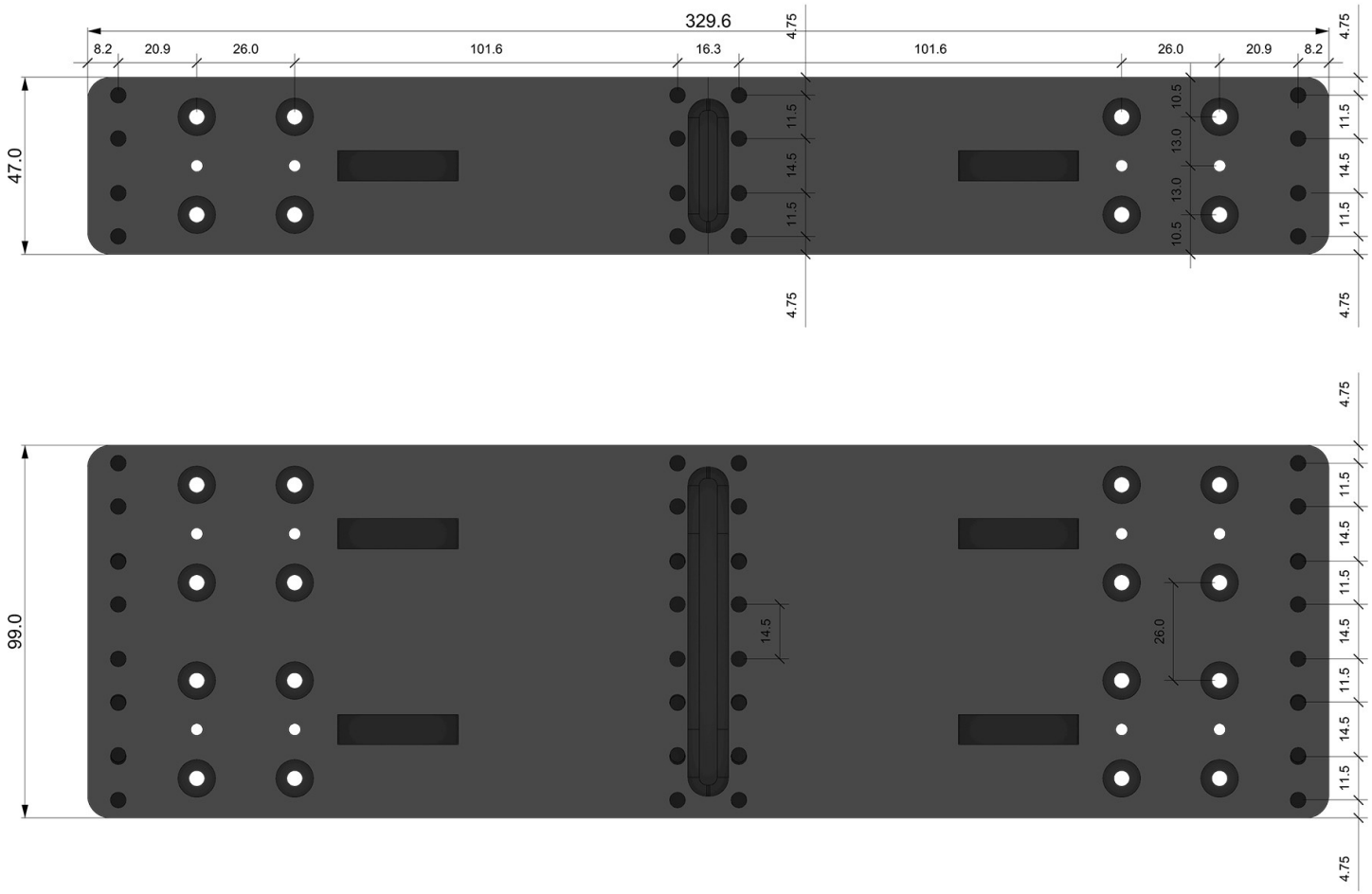
18. Cut 2 M16 rods to height of tower and slide with handle; 19. Drill M6 holes & tap rods to attach handle.

20



20. Attach IKEA or USSA pegboards on backbones (official bench accessories at ussadesign.com).

1F & 2F L Backbone Dimensions



Casters Mode





Flip the base, insert 4 casters with M8 bolts and flip it back for full mobility.